The Singular Value Decomposition and Norms

Math 5250 — Week 7

Overview

- The singular value decomposition (SVD): a version of the spectral theorem for non-square matrices.
- Low rank approximation, with an application to image compression.
- Norms on vector spaces; all norms on $\mathbf{R}^n$ or $\mathbf{C}^n$ are equivalent.
- The operator norm on matrices.

The Singular Value Decomposition

Setup

Let X be a real $n \times m$ matrix and let $D = X^\top X$.

- D is a symmetric, positive semidefinite $m \times m$ matrix.
- By the spectral theorem there is an orthogonal P with

$$P^\top D P = \Lambda, \qquad \text{equivalently} \qquad DP = P\Lambda,$$

so the columns of P are orthonormal eigenvectors for D.

Set $A = XP$. Then

$$A^\top A = P^\top X^\top X P = \Lambda,$$

so the columns of A are orthogonal, but not necessarily orthonormal.

Geometrically, row i of A gives the coordinates of row i of X in the orthonormal basis given by the columns of P.

Normalizing the Columns

Full rank case. If D is positive definite (all $\lambda_i > 0$), which happens exactly when X has rank $m \leq n$, set

$$A_1 = A\Lambda^{-1/2}.$$

Then $A_1^\top A_1 = \Lambda^{-1/2}\Lambda\Lambda^{-1/2} = I$, so A_1 has orthonormal columns.

General case. Order the eigenvalues $\lambda_1 \geq \cdots \geq \lambda_m$, with $\lambda_{r+1} = \cdots = \lambda_m = 0$, so X has rank r.

- Columns $r+1, \ldots, m$ of A are zero; drop them to get A_r.
- Shrink Λ to the $r \times r$ diagonal matrix Λ_r of nonzero eigenvalues.
- $A_1 = A_r\Lambda_r^{-1/2}$ is $n \times r$ with orthonormal columns.

Building U and Σ

- Complete the columns of A_1 to an orthogonal $n \times n$ matrix U by adding orthonormal columns $u_{r+1}, \ldots, u_n$.
- Let $\sigma_i = \sqrt{\lambda_i}$ and let Σ be the $n \times m$ "diagonal" matrix of these *singular values*.

In $U\Sigma$:

- the first r columns are $A_1 \Lambda_r^{1/2}$, which are the first r columns of $A = XP$;
- the remaining columns are zero, as are the remaining columns of XP.

The SVD

Theorem. Therefore $XP = U\Sigma$, or

$$X = U\Sigma V^\top$$

where U $(n \times n)$ and $V = P$ $(m \times m)$ are orthogonal and Σ is $n \times m$ "diagonal."

The singular values (the diagonal entries of Σ) are uniquely determined.

Reduced SVD. Keep the first r columns of U (U_r, $n \times r$), the $r \times r$ block Σ_r, and the first r columns of V (V_r, $m \times r$):

$$X = U_r \Sigma_r V_r^\top.$$

Low Rank Approximation

A Sum of Rank One Maps

If X is the matrix of $L : \mathbf{R}^m \to \mathbf{R}^n$, the reduced SVD says

$$X = \sum_{i=1}^{r} \sigma_i u_i v_i^\top,$$

where u_i, v_i are the columns of U, V.

- $u_i v_i^\top$ is an $(n \times 1)(1 \times m)$ product, an $n \times m$ matrix.
- It has rank one: its range is spanned by u_i.

Truncating the Sum

$$X_k = \sum_{i=1}^{k} \sigma_i u_i v_i^\top$$

- If $\sigma_i > 0$ for $i \leq k$, then X_k has rank k.
- We will show X_k is the *best* approximation to X among matrices of rank at most k.
- There are several meanings of "best"; the most useful in applications: X_k minimizes the squared error

$$E(B, X) = \sum_{i,j} (x_{ij} - b_{ij})^2$$

over all matrices $B = (b_{ij})$ of rank at most k.

Application: Image Compression

- A gray-scale image is an array of integers, one per pixel (0 for black, 255 for white).
- A color image is three such arrays (red, green, blue).

Let A be an $n \times n$ gray-scale image.

- As k increases, A_k gets close to A, so it represents "nearly" the same image.
- Storing A takes n^2 numbers.
- Storing A_k takes k columns of U, k columns of V, and k singular values:

$$2kn + k = k(2n + 1) \text{ numbers.}$$

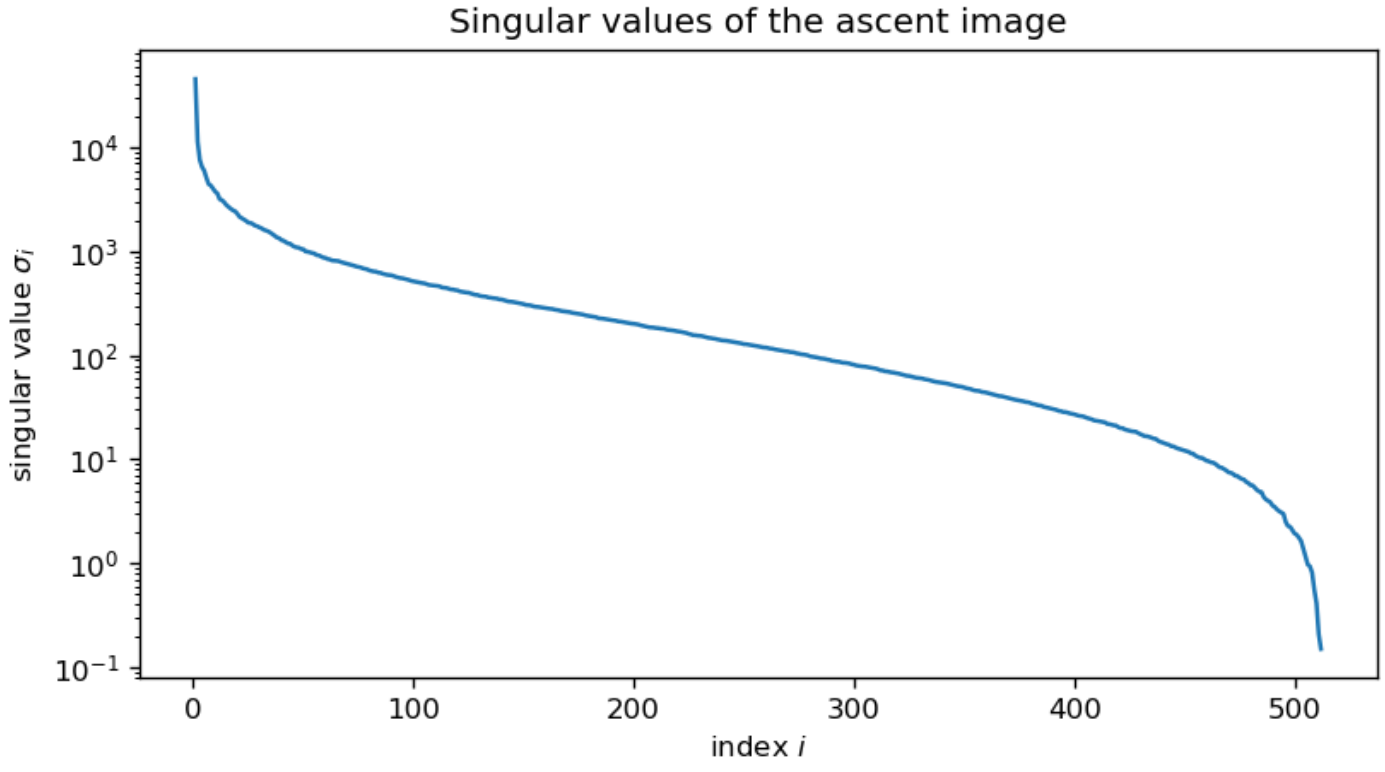
Example: `scipy.datasets.ascent()`

A 512×512 gray-scale image: a matrix A of integers between 0 and 255.



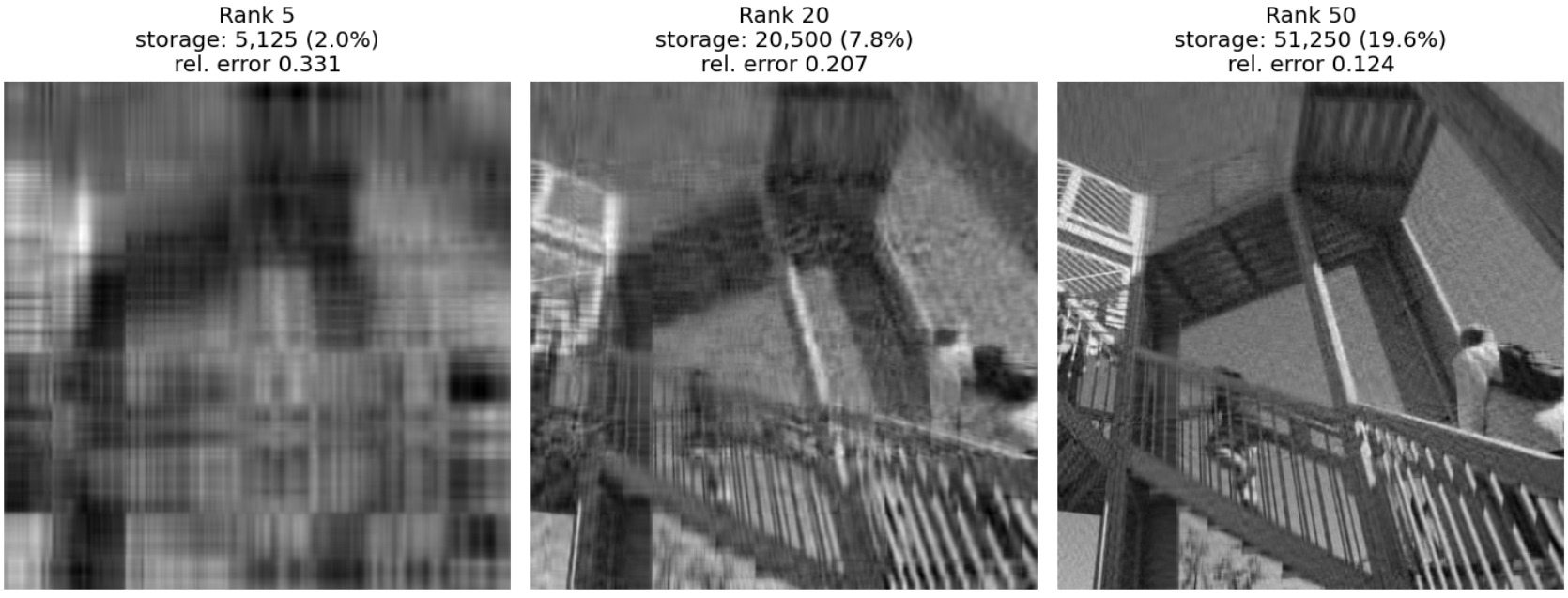
Singular Values of the Image

The largest singular value is about 45500; the smallest is about 0.15.



A relatively small number larger than 10^3; many between 10^3 and 1; a few less than 1.

Low Rank Approximations



rank k	storage $k(2n+1)$	% of $512^2 = 262{,}144$	relative error
5	5,125	2.0%	0.331
20	20,500	7.8%	0.207
50	51,250	19.6%	0.124

Norms

Norms on Vector Spaces

Let V be a vector space over $F = \mathbf{R}$ or $\mathbf{C}$. A **norm** on V is a function $\|\cdot\| : V \to \mathbf{R}$ such that:

- $\|v\| \geq 0$, with equality if and only if $v = 0$;
- $\|v + w\| \leq \|v\| + \|w\|$;
- $\|cv\| = |c|\|v\|$ for $c \in F$.

Dropping "$\|v\| = 0$ only if $v = 0$" gives a **seminorm**.

A norm makes V a metric space with $d(v, w) = \|v - w\|$.

Two Key Examples

The sup norm relative to a basis $e_1, \ldots, e_n$:

$$\| \sum_{i=1}^{n} a_i e_i \|_{\sup} = \max\{|a_1|, \ldots, |a_n|\}.$$

The ℓ^2 (Euclidean) norm from an inner product:

$$\|v\| = \sqrt{\langle v, v \rangle}.$$

For the usual dot product, this gives Euclidean distance.

The Triangle Inequality for ℓ^2

Cauchy–Schwarz: $|\langle v, w\rangle| \le \sqrt{\langle v, v\rangle\langle w, w\rangle}$. So

$$\begin{aligned}\|v + w\|^2 &= \langle v, v\rangle + 2\,\mathrm{Re}\langle v, w\rangle + \langle w, w\rangle \\ &\le \|v\|^2 + 2|\langle v, w\rangle| + \|w\|^2 \\ &\le \|v\|^2 + 2\|v\|\|w\| + \|w\|^2 = (\|v\| + \|w\|)^2.\end{aligned}$$

Take square roots.

Equivalent Norms

Definition. Norms $\|\cdot\|_1$ and $\|\cdot\|_2$ on V are **equivalent** if there are constants $A, B > 0$ with

$$A\|v\|_1 \leq \|v\|_2 \leq B\|v\|_1 \quad \text{for all } v \in V.$$

Proposition. All norms on $V = \mathbf{R}^n$ or $\mathbf{C}^n$ are equivalent.

Strategy: show any norm ρ is equivalent to the sup norm for the standard basis.

The Upper Bound

Let ρ be any norm and $v = \sum_{i=1}^{n} a_i e_i$. Then

$$\begin{aligned}\rho(v) &\le \sum_{i=1}^{n} |a_i| \rho(e_i) \\ &\le \Big(\sum_{i=1}^{n} \rho(e_i)\Big) \|v\|_{\sup}.\end{aligned}$$

So $\rho(v) \le B\|v\|_{\sup}$ with $B = \sum_i \rho(e_i)$.

Lemma. ρ is continuous for the metric given by $\|\cdot\|_{\sup}$:

$$|\rho(x) - \rho(y)| \le \rho(x - y) \le B\|x - y\|_{\sup}.$$

This much works for seminorms too. The next step needs $\rho(v) = 0 \Rightarrow v = 0$.

Compactness of the Sphere

Lemma. The unit sphere $S = \{v : \|v\|_{\sup} = 1\}$ is compact.

Proof.

- The closed ball $\{\|v\|_{\sup} \leq 1\}$ is a finite product of intervals $[-1, 1]$ (or closed disks $|z| \leq 1$), so it is compact.
- The open ball $\{\|v\|_{\sup} < 1\}$ is a finite product of open intervals (or open disks), so it is open.
- S is the closed ball minus the open ball, so it is closed, hence compact. $\square$

The Lower Bound

ρ is continuous on the compact set S, so it attains a minimum A, and $A > 0$ since ρ is nonzero on S.

For $v \neq 0$:

$$\rho\Big(\frac{v}{\|v\|_{\sup}}\Big) \geq A \quad \Longrightarrow \quad \rho(v) \geq A\|v\|_{\sup}.$$

Altogether:

$$A\|v\|_{\sup} \leq \rho(v) \leq B\|v\|_{\sup}. \qquad \square$$

Remarks and Consequences

- The key input is compactness of closed intervals in $\mathbf{R}$ or closed disks in $\mathbf{C}$ (Heine–Borel).
- Conrad, Equivalence of Norms: all norms on a finite dimensional vector space over a complete valued field are equivalent.

Corollary. For finite dimensional V, W over $\mathbf{R}$ or $\mathbf{C}$:

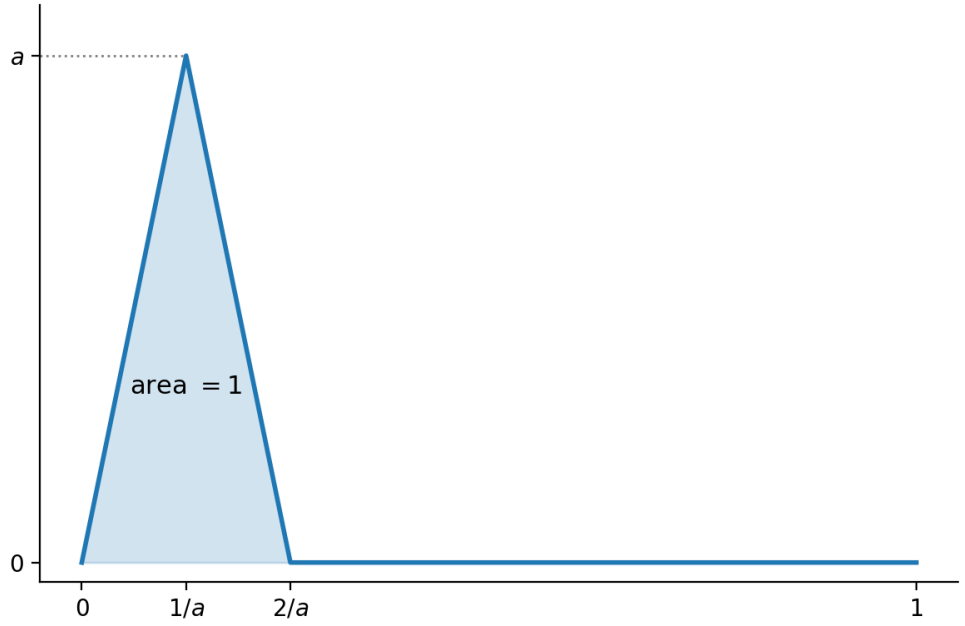
- every linear map $V \to W$ is continuous in any norm topology;
- bounded, open, closed and compact sets are the same for every norm.

Infinite Dimensions Are Different

On $C([0,1])$ consider

$$\|f\|_{\sup} = \sup_{0 \le x \le 1} |f(x)|, \qquad \|f\|_{L^1} = \int_0^1 |f(x)|\, dx.$$

A spike of height a and area 1 has $\|f\|_{\sup} = a$ and $\|f\|_{L^1} = 1$, so these norms are not equivalent.



Matrix Norms

Matrices as Vectors

$M_{n\times m}(F) \cong F^{nm}$, so matrices have a sup norm $\|\cdot\|_{\sup}$ and an ℓ^2 norm, and all norms on matrices are equivalent.

For $A \in M_n(F)$ let

$$\rho_A(v) = \|Av\| \qquad (\ell^2 \text{ norm on } F^n).$$

ρ_A is a seminorm:

- $\rho_A(v) \geq 0$;
- $\rho_A(v+w) = \|Av + Aw\| \leq \rho_A(v) + \rho_A(w)$;
- $\rho_A(av) = |a|\rho_A(v)$.

So ρ_A is continuous, and $\rho_A(v) \leq C\|v\|$ for some $C \geq 0$.

The Operator Norm

Proposition. There is a smallest $b \geq 0$ with $\rho_A(v) \leq b\|v\|$ for all v.

Proof. ρ_A is continuous on the compact sphere $\{\|v\| = 1\}$, so it attains a maximum b. $\square$

Definition. The **operator norm** of A is

$$\|A\| = \max_{\|v\|=1} \|Av\|.$$

Properties of the Operator Norm

Proposition. The operator norm is a norm on $M_n(F)$, and for $A, B \in M_n(F)$:

- $\|AB\| \leq \|A\|\|B\|$;
- $\|A^*\| = \|A\|$;
- $\|A^*A\| = \|AA^*\| = \|A\|^2$;
- $|\langle Av, w\rangle| \leq \|A\|\|v\|\|w\|$;
- $\|A\|_{\sup} \leq \|A\| \leq n\sqrt{n}\,\|A\|_{\sup}$.

Here $A^* = A^\top$ when $F = \mathbf{R}$.

So the operator norm gives $M_n(F)$ the usual topology of F^{n^2}, and $M_n(F)$ is complete for it.

(Conrad, Computing the Norm of a Matrix, Theorem 3.4.)

Proofs

Products. $\|ABv\| \le \|A\|\|Bv\| \le \|A\|\|B\|\|v\|$, and $\|AB\|$ is the smallest such constant.

Inner products. $|\langle Av, w\rangle| \le \|Av\|\|w\| \le \|A\|\|v\|\|w\|$ by Cauchy–Schwarz.

Proof: the Adjoint

By Cauchy–Schwarz,

$$\|Av\|^2 = \langle Av, Av\rangle = \langle v, A^*Av\rangle \leq \|v\|\|A^*Av\|,$$

so

$$\|A\|^2 \leq \|A^*A\| \leq \|A^*\|\|A\|.$$

- If $A \neq 0$, divide by $\|A\|$: $\|A\| \leq \|A^*\|$.
- Apply this to A^*: $\|A^*\| \leq \|A\|$. So they are equal.
- Then $\|A\|^2 \leq \|A^*A\| \leq \|A\|^2$, so $\|A^*A\| = \|A\|^2$; similarly for AA^*. $\square$